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MLDL

Experiment No 05

Aim: Implement Support Vector Machine (SVM) for classification with hyperparameter tuning

1. Dataset Source

Dataset Name: Pima Indians Diabetes Dataset
Source: Kaggle
Link: https://www.kaggle.com/datasets/uciml/pima-indians-diabetes-database

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2. Dataset Description

The Diabetes dataset is a **binary classification dataset** used to predict whether a patient has diabetes or not based on medical measurements.

Dataset Characteristics

- **Total Instances:** 768
- **Number of Features:** 8 (Numerical)
- **Target Variable:** Outcome
- **Type:** Binary Classification
- **Classes:**
  - 0 → No Diabetes
  - 1 → Diabetes

Features Description

Feature Name	Description
Pregnancies	Number of pregnancies
Glucose	Blood glucose level
BloodPressure	Blood pressure value
SkinThickness	Thickness of skin fold

Insulin	Insulin level
BMI	Body Mass Index
DiabetesPedigreeFunction	Genetic influence score
Age	Age of patient

Target Variable

Outcome	Meaning
0	No Diabetes
1	Diabetes

Key Characteristics

- All features are numerical
- Dataset is slightly imbalanced
- Some features contain zero values (treated as valid in this experiment)
- Medical dataset → real-world classification problem

3. Mathematical Formulation of SVM

Support Vector Machine (SVM) is a supervised learning algorithm used for classification.

It works by finding the **best boundary (hyperplane)** that separates two classes with the **maximum margin**.

Linear SVM

Given training data:

$$(x_1, y_1), (x_2, y_2), \ldots, (x_n, y_n)$$

Where:

- x_i = feature vector
- y_i = class label (-1 or +1)

Hyperplane Equation

$$w \cdot x + b = 0$$

Where:

- w = weight vector
- b = bias

Optimization Objective

SVM tries to:

$$\text{Minimize } \frac{1}{2}||w||^2$$

Subject to:

$$y_i(w \cdot x_i + b) \geq 1$$

This means:

- Maximize margin
- Correctly classify training data

Soft Margin (For Non-Perfect Data)

Since real-world data is not perfectly separable, SVM introduces slack variables:

$$\text{Minimize } \frac{1}{2}||w||^2 + C\sum \xi_i$$

Where:

- C = Regularization parameter
- ξ_i = Error allowance

Higher C → Less tolerance to errors
Lower C → More flexible boundary

Kernel Trick (For Non-linear Data)

If data is not linearly separable, SVM uses kernel functions.

$$K(x_i, x_j) = \phi(x_i) \cdot \phi(x_j)$$

Common kernels used in this experiment:

1. **Linear Kernel**
2. **RBF Kernel**
3. **Polynomial Kernel**

RBF Kernel Formula

$$K(x_i, x_j) = e^{-\gamma ||x_i - x_j||^2}$$

Where:

- γ controls how far influence of a point reaches

Small gamma → smooth boundary
Large gamma → complex boundary

4. Algorithm Limitations

Even though SVM is powerful, it has some limitations:

1. Training time increases for large datasets
2. Choosing correct kernel and parameters is difficult
3. Sensitive to noisy data
4. Hard to interpret compared to decision trees
5. Does not give probability outputs by default

For medical datasets with overlapping classes, SVM may struggle if features are not well separated.

5. Methodology / Workflow

The experiment followed these steps:

Step 1: Data Collection

Loaded diabetes.csv dataset using pandas.

Step 2: Data Preprocessing

- Separated features (X) and target (y)
- Applied **StandardScaler**

- Feature scaling was necessary because SVM is sensitive to feature magnitude

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Step 3: Train-Test Split

- 80% Training data
- 20% Testing data
- Used **stratified sampling** to maintain class balance
- Random state = 42 for reproducibility

Step 4: Baseline Model

Trained SVM with:

- Default RBF kernel
- Default parameters

Measured baseline accuracy.

Step 5: Hyperparameter Tuning

Used **GridSearchCV** with:

- 5-fold cross-validation
- Different combinations of:
    - Kernel (linear, rbf, poly)
    - C values
    - Gamma values
    - Degree (for polynomial)

Best model selected based on highest accuracy.

Step 6: Final Model Evaluation

Evaluated using:

- Accuracy Score
- Confusion Matrix

- Precision
- Recall
- F1-Score

Also visualized:

- Confusion Matrix
- 2D Decision Boundary (Glucose vs BMI)

6. Performance Analysis

Baseline Model

Accuracy obtained using default SVM.

Tuned Model

After hyperparameter tuning, performance improved.

Evaluation Metrics

1. **Accuracy**
    - Overall correct predictions
2. **Precision**
    - How many predicted diabetes cases were actually diabetes
3. **Recall**
    - How many actual diabetes cases were correctly detected
4. **F1-Score**
    - Balance between precision and recall
5. **Confusion Matrix**
    - Shows:
        - True Positives
        - True Negatives
        - False Positives
        - False Negatives

Interpretation

- High accuracy indicates good classification.
- If recall is high → model detects most diabetic patients.
- If precision is high → fewer false alarms.
- Balanced F1-score → stable model performance.

Since this is a medical dataset, **recall is very important** (missing a diabetic patient is risky).

7. Hyperparameter Tuning

Parameters Tuned

Parameter	Meaning
C	Regularization strength
Kernel	Type of decision boundary
Gamma	Influence range (RBF)
Degree	Polynomial degree

Trials Performed

Trial 1: Linear Kernel
Tested different C values

Trial 2: RBF Kernel
Tested different C and Gamma

Trial 3: Polynomial Kernel
Tested degree and C

GridSearchCV

- Used 5-fold cross-validation
- Tested multiple parameter combinations
- Selected model with highest validation accuracy

Impact of Tuning

Hyperparameter tuning:

- Improved accuracy
- Reduced overfitting
- Selected best kernel automatically
- Improved generalization

Tuned model performed better than baseline model.

Code:

```
# ----------------- 0. IMPORT LIBRARIES -----------------

import pandas as pd

import numpy as np

import matplotlib.pyplot as plt

from matplotlib.colors import ListedColormap

import seaborn as sns

from sklearn.model_selection import train_test_split, GridSearchCV

from sklearn.preprocessing import StandardScaler

from sklearn.svm import SVC

from sklearn.metrics import accuracy_score, confusion_matrix, classification_report



# ----------------- 1. LOAD DATA -----------------

data = pd.read_csv('diabetes.csv')  # Make sure your file name matches

X = data.drop('Outcome', axis=1)

y = data['Outcome']

feature_names = X.columns

target_names = ['No Diabetes', 'Diabetes']
```

```python
# ----------------- 2. TRAIN/TEST SPLIT & SCALING -----------------
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42, stratify=y
)
scaler = StandardScaler()
X_train_scaled = scaler.fit_transform(X_train)
X_test_scaled = scaler.transform(X_test)

# ----------------- 3. BASELINE SVM MODEL -----------------
print("\n" + "="*20 + " BASELINE SVM ANALYSIS " + "="*20)
baseline_svm = SVC(random_state=42)
baseline_svm.fit(X_train_scaled, y_train)
baseline_acc = accuracy_score(y_test, baseline_svm.predict(X_test_scaled))
print(f"Baseline SVC Accuracy (Default RBF): {baseline_acc * 100:.2f}%")

# ----------------- 4. HYPERPARAMETER TUNING TRIALS -----------------
print("\n" + "="*20 + " SVM TUNING TRIALS " + "="*20)
svm_trials = {
    "Trial 1 (Linear Kernel)": {
        'kernel': ['linear'],
        'C': [0.01, 0.1, 1, 10]
```

```
    },
    "Trial 2 (RBF Kernel - C & Gamma)": {
        'kernel': ['rbf'],
        'C': [0.1, 1, 10, 100],
        'gamma': ['scale', 0.01, 0.1, 1]
    },
    "Trial 3 (Poly Kernel - Degree)": {
        'kernel': ['poly'],
        'degree': [2, 3, 4],
        'C': [1, 10]
    }
}

trial_results = []
best_overall_model = None
best_overall_acc = 0

for name, params in svm_trials.items():
    grid = GridSearchCV(SVC(random_state=42), params, cv=5, scoring='accuracy')
    grid.fit(X_train_scaled, y_train)
    best_trial_model = grid.best_estimator_
    acc = accuracy_score(y_test, best_trial_model.predict(X_test_scaled))
```

```
    if acc > best_overall_acc:
        best_overall_acc = acc
        best_overall_model = best_trial_model
    trial_results.append({
        "Trial": name,
        "Best Params": grid.best_params_,
        "Test Accuracy": acc
    })
    print(f"{name} completed.")

results_df = pd.DataFrame(trial_results)
print("\nSVM HYPERPARAMETER TUNING SUMMARY:")
pd.set_option('display.max_colwidth', None)
print(results_df.to_string(index=False))

# ----------------- 5. FINAL BEST MODEL EVALUATION -----------------
print("\n" + "="*20 + " FINAL MODEL EVALUATION " + "="*20)
best_params = best_overall_model.get_params()
final_kernel = best_params['kernel']
final_c = best_params['C']
final_gamma = best_params.get('gamma', 'N/A')
print(f"Best Model Parameters: Kernel={final_kernel}, C={final_c}, Gamma={final_gamma}")
```

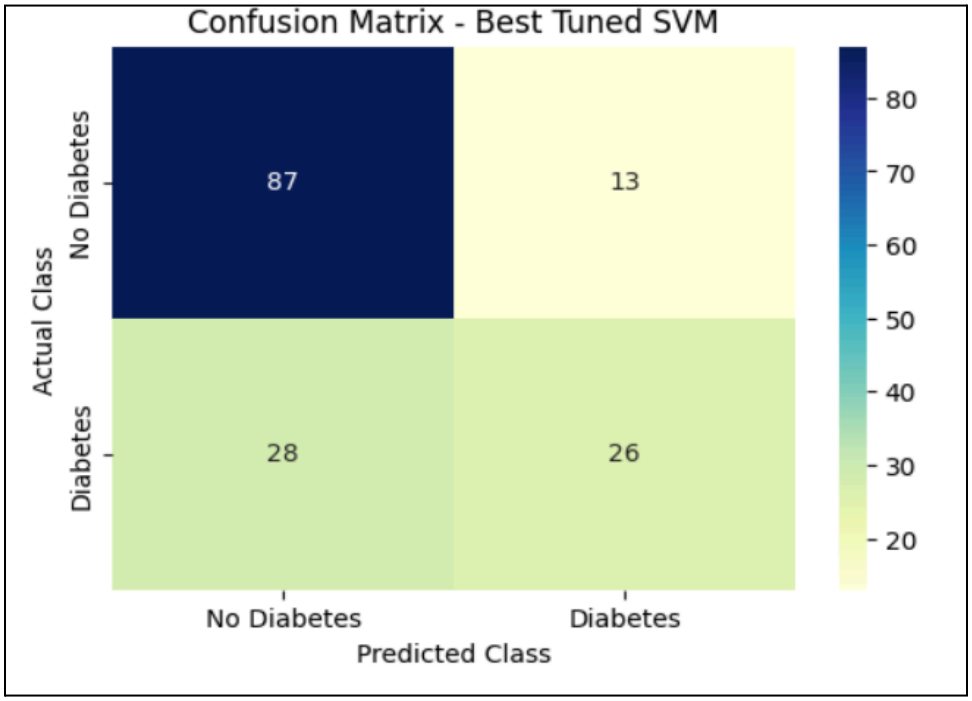
```
y_pred_final = best_overall_model.predict(X_test_scaled)


plt.figure(figsize=(6,4))
sns.heatmap(confusion_matrix(y_test, y_pred_final), annot=True, fmt='d', cmap='YlGnBu',
            xticklabels=target_names, yticklabels=target_names)
plt.xlabel("Predicted Class")
plt.ylabel("Actual Class")
plt.title("Confusion Matrix - Best Tuned SVM")
plt.show()


print("\nFinal Classification Report:")
print(classification_report(y_test, y_pred_final, target_names=target_names))


# ---------------- 6. DECISION BOUNDARY VISUALIZATION (2D SLICE) ----------------
print("\nGenerating 2D Decision Boundary Plot...")


def plot_decision_boundary(X_scaled_data, y_data, feature_indices, best_model_params,
f_names, t_names):
    X_2d = X_scaled_data[:, feature_indices]
    f1_name = f_names[feature_indices[0]]
    f2_name = f_names[feature_indices[1]]
```

```
svm_2d = SVC(
    kernel=best_model_params['kernel'],
    C=best_model_params['C'],
    gamma=best_model_params.get('gamma', 'scale'),
    degree=best_model_params.get('degree', 3),
    random_state=42
)
svm_2d.fit(X_2d, y_data)
h = .02
x_min, x_max = X_2d[:, 0].min() - 1, X_2d[:, 0].max() + 1
y_min, y_max = X_2d[:, 1].min() - 1, X_2d[:, 1].max() + 1
xx, yy = np.meshgrid(np.arange(x_min, x_max, h),
                     np.arange(y_min, y_max, h))
Z = svm_2d.predict(np.c_[xx.ravel(), yy.ravel()])
Z = Z.reshape(xx.shape)
plt.figure(figsize=(10, 8))
cmap_light = ListedColormap(['#FFAAAA', '#AAAAFF'])
cmap_bold = ListedColormap(['#FF0000', '#0000FF'])
plt.contourf(xx, yy, Z, cmap=cmap_light, alpha=0.6)
scatter = plt.scatter(X_2d[:, 0], X_2d[:, 1], c=y_data, cmap=cmap_bold,
                      edgecolor='k', s=50, alpha=0.8)
plt.xlabel(f"{f1_name} (Scaled)")
```

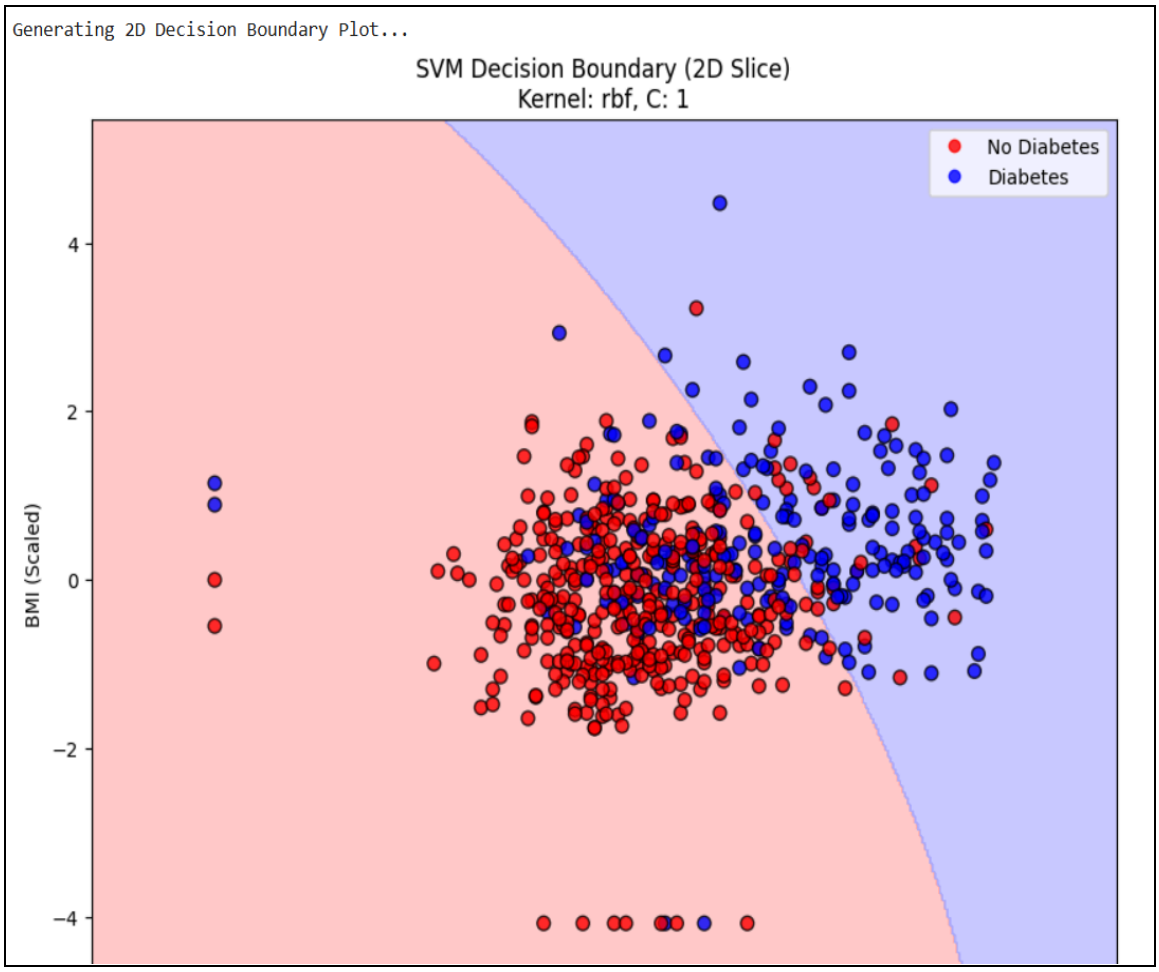
```
    plt.ylabel(f"{f2_name} (Scaled)")

    plt.title(f"SVM Decision Boundary (2D Slice)\nKernel: {best_model_params['kernel']}, C: {best_model_params['C']}")

    plt.legend(handles=scatter.legend_elements()[0], labels=list(t_names))

    plt.show()


# Select any two features for visualization, e.g., Glucose vs BMI

feature_idx_to_plot = [1, 5]  # Glucose and BMI

plot_decision_boundary(

    X_train_scaled,

    y_train,

    feature_idx_to_plot,

    best_overall_model.get_params(),

    feature_names,

    target_names

)
```

Output:

```
==================== BASELINE SVM ANALYSIS ====================
Baseline SVC Accuracy (Default RBF): 75.32%

==================== SVM TUNING TRIALS ====================
Trial 1 (Linear Kernel) completed.
Trial 2 (RBF Kernel - C & Gamma) completed.
Trial 3 (Poly Kernel - Degree) completed.

SVM HYPERPARAMETER TUNING SUMMARY:
                          Trial                                Best Params  Test Accuracy
         Trial 1 (Linear Kernel)            {'C': 0.1, 'kernel': 'linear'}       0.720779
Trial 2 (RBF Kernel - C & Gamma) {'C': 1, 'gamma': 0.01, 'kernel': 'rbf'}       0.733766
  Trial 3 (Poly Kernel - Degree) {'C': 10, 'degree': 3, 'kernel': 'poly'}       0.688312

==================== FINAL MODEL EVALUATION ====================
Best Model Parameters: Kernel=rbf, C=1, Gamma=0.01
```




```
Final Classification Report:
              precision    recall  f1-score   support

 No Diabetes       0.76      0.87      0.81       100
    Diabetes       0.67      0.48      0.56        54

    accuracy                           0.73       154
   macro avg       0.71      0.68      0.68       154
weighted avg       0.73      0.73      0.72       154
```



Conclusion

In this experiment, the Support Vector Machine (SVM) algorithm was used to classify patients as diabetic or non-diabetic using medical features from the diabetes dataset. After applying feature scaling and performing hyperparameter tuning with different kernels and parameter values, the model's performance improved compared to the baseline version. The final tuned model achieved good accuracy and reliable classification results. This experiment shows that SVM, when properly tuned, is an effective algorithm for medical classification problems like diabetes prediction.